

Hyeongmin Choi

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Education

Chosun University

B.S. in Artificial Intelligence Engineering (AI/SW School)

Gwangju, South Korea
Mar. 2023 - Present (2nd Year)

- **Academic Details:** 3 years enrollment period (including military leave)
- **Relevant Coursework:** Artificial Intelligence, C++ Programming, Introduction to OSS (Open Source Software)

Skills

Core Competencies: C++ Programming, Data Structures, Real-Time Data Ingestion
AI-Assisted Development: Prompt Engineering, AI Agent Pairing, AI-Driven Prototyping (Vibe Coding)
Web Automation & Database: Flask Backend, Selenium Automation, Pandas Data Wrangling, SQLite/MySQL
Tools & Infrastructure: Docker, Docker Compose, Git, GitHub, Linux Environments

Projects

ASC_TLMSYS (Motorsport Telemetry & Analysis System)

Sep. 2025 - Present

Role: Sole Developer (Personal Project)

- Developed a real-time 'F1 Pit Wall Control' telemetry dashboard serving dynamic vehicle physics data over a FastAPI (Port 8000) backend.
- Built a high-performance UDP socket listener (Port 8001) in Python/C++ to ingest live simulator packet streams at zero lag.
- Engineered live analytics tracking tire core/IMO temps (81.7°C), brake temps (417°C), suspension load, ERS battery, and DRS lock states.
- **Future Roadmap:** Scaling to hardware telemetry, designing sensor-acquisition boards on student-built formula race cars to ingest real track telemetry.

SCSS (Self Check-in Integrated System) v2.0

Jan. 2025 - Apr. 2026

Role: Sole Developer / System Architect (Built during military service)

- Developed and operationalized a 24/7 unmanned self check-in kiosk interface on a Windows OS PC configured in Kiosk Mode.
- **Self-Taught & AI Collaboration:** Self-taught Python and network architectures, partnering with an AI agent to architect the program while serving as a Private (■■■).
- Engineered an automated backend parser (Pandas & APScheduler) executing daily at 14:50 to process portal reservation Excel logs.
- Constructed direct database queries mapping Guest Reservation Numbers to dynamic Doorlock Passwords with Uvicorn (Port 80) serving requests.
- Containerized the software stacks using Docker Compose, guaranteeing environment isolation and zero-downtime operations.

Experience

System Administrator & Automation Developer

Jan. 2025 - Apr. 2026

Republic of Korea Army (Sangseung Restel IT Automation)

- Pioneered manual administrative work automation in the unit, decreasing checkout processing bottlenecks by 90%.
- Established local intranet routing linking user kiosk client terminals with isolated dockerized backend database server.

Telemetry Infrastructure Lead

Mar. 2024 - Present

Chosun University Formula SAE (Student Race Car Team)

- Joined to bridge mechanical engineering & embedded systems with AI/SW components, focusing on vehicle dynamics and telemetrics.
- Identified data logging instability on physical telemetry hardware and designed C++ real-time packet filtering modules.
- Implemented low-latency data pipelines for sensor preprocessing, shifting unstable legacy structures to stable telemetry platforms.

Career Goal

Aspiring **Race Data Engineer**. Dedicated to leveraging solid foundations in C++, low-latency telemetry data structures, and real-time sensor ingestion pipelines to optimize race car setups, track dynamics, and race strategies.